

First Negative Degrees in Integer Thue–Morse Pressure

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Abstract

Let N_m be the first even degree strictly above $2m$ at which the phase Taylor coefficient of integer Thue–Morse pressure is negative. We prove $N_m = \gamma m - \Lambda^{-1} \log \log(2m) + O(1)$, where a positive-series saddle equation defines $\gamma \in (6.662966, 6.662968)$ and $\Lambda > 0$. In particular N_m/m converges to a value strictly below $20/3$. The key comparison is between a positive first model and an actual quadratic Schur term with multiplier $2 \log_2(2m) + O(1)$. A principal inverse-branch identity bounds the frozen correction at the second-model scale. Exact rational compact-domain certificates join the local saddle arcs to the rest of the coefficient contour. The result is asymptotic; it does not supply an unconditional floor rule or an effective onset threshold.

Status. This is unrefereed ordinary mathematics with exact rational interval certificates, not a Lean formalization. Earlier analytic prerequisites are included unchanged with the sources. No exhaustive priority claim is made.

1 Statement and the two positive models

Use the cosine-phase normalization of the repository manuscript [1], and put

$$P_m(t) = p_m(t/\pi), \quad d = 2m, \quad N_m = \min\{N > d : N \text{ even, } [t^N]P_m(t) < 0\}.$$

The infinite-sign theorem [4] ensures that N_m exists for every integer $m \geq 2$. The lower negative coefficients and the cancellation at degree $2m$ are excluded from this minimum.

Define

$$c = 2/\pi, \quad b_j = \tan(\pi/2^{j+1}), \quad B(a) = \prod_{j \geq 1} (1 + ab_j), \quad L(a) = cB(a), \quad \mathcal{G}(t) = \tan t L(\tan t).$$

The product B is entire. The series $\mathcal{G}$ has strictly positive coefficients at every degree at least one and is analytic on $|t| < \pi/2$. Put

$$\mu(t) = \frac{t\mathcal{G}'(t)}{\mathcal{G}(t)}.$$

For $\sigma > k$, let $t_k(\sigma) \in (0, \pi/2)$ solve $\mu(t_k) = \sigma/k$, and set

$$\Phi_k(\sigma) = k \log \mathcal{G}(t_k) - \sigma \log t_k. \tag{1}$$

The logarithmic derivative is the mean of a positive coefficient law; its derivative in $\log t$ is its strictly positive variance. Moreover μ increases from one to infinity, so each saddle exists and is unique.

There is a unique crossing $\sigma_* > 2$ of Φ_1 and Φ_2 . Indeed

$$\Delta(\sigma) := \Phi_2(\sigma) - \Phi_1(\sigma), \quad \Delta'(\sigma) = \log(t_1(\sigma)/t_2(\sigma)) > 0, \quad (2)$$

and the rational certificate gives opposite signs at 3.331483 and 3.331484. Write

$$\gamma = 2\sigma_*, \quad \Lambda = \log \frac{t_1(\sigma_*)}{t_2(\sigma_*)}.$$

Numerical orientation gives $\gamma \approx 6.6629666581$ and $\Lambda \approx 0.8062663610$; the rigorous slope interval used here is the one above.

Theorem 1.1. *As the integer m tends to infinity,*

$$N_m = \gamma m - \frac{\log \log(2m)}{\Lambda} + O(1). \quad (3)$$

In particular $N_m/m \rightarrow \gamma \in (6.662966, 6.662968)$.

All unqualified logarithms are natural. The bounded term includes the even-degree lattice. No unconditional rounding formula is asserted.

Related context. Fan, Schmeling and Shen [5] study the same phase-dependent cosine-product family through ergodic optimization and multifractal analysis. Baake, Gohlke, Kesseböhmer and Schindler [6] study pressure and scaling for the classical Thue–Morse measure. The variable and asymptotic regime here concern high-order phase Taylor signs as the integer moment order grows. These references provide context, rather than an exhaustive novelty determination.

2 Schur terms and the analytic prerequisites

The normalized transfer operator is

$$(V_a f)(x) = \sum_{2y=x \pmod{1}} (\cos \pi y + a \sin \pi y)^d f(y).$$

Let $\lambda(a)$ be the eigenvalue branch of V_a with $\lambda(0) = 1$. The pressure germ is exactly

$$P_m(t) = d \log \cos t + \log \lambda(\tan t),$$

with the logarithm zero at the origin. Let h_0 be the fixed function of V_0 , normalized by $h_0(0) = 1$, let $\text{ev } f = f(1/2)$, and put $Qf = f - f(0)h_0$. On functions vanishing at zero set

$$\begin{aligned} \mathcal{B}_a &= QV_a Q, & q_a &= QV_a h_0, & F &= \text{ev } h_0 + \text{ev}(I - \mathcal{B}_a)^{-1} q_a, \\ G &= \text{ev}(I - \mathcal{B}_a)^{-2} q_a, & H &= \text{ev}(I - \mathcal{B}_a)^{-3} q_a. \end{aligned}$$

The functions are even analytic germs, with $G(0) = H(0) = 0$. If $\delta = \lambda - 1$ for the normalized simple eigenvalue, then

$$\delta = a^d \{F - \delta G + \delta^2 H + O(\delta^3)\}.$$

Consequently, below degree $4d$,

$$\begin{aligned} P_m(t) &= d \log \cos t + \tan^d t F(\tan t) - \tan^{2d} t K_2(\tan t) + \tan^{3d} t K_3(\tan t), \\ K_2 &= FG + F^2/2, & K_3 &= FG^2 + F^2 H + F^2 G + F^3/3. \end{aligned} \quad (4)$$

This is an exact identity of Taylor coefficients.

We use the preceding all-order and eventual-6.6m reports [2, 3]. They are bundled unchanged, with their reproducible sources. The needed facts are stated explicitly here.

First, $[t^N]P_m > 0$ for every $m \geq 2$ and even $d < N < 3d$; moreover, for all sufficiently large m , this holds throughout $d < N \leq 3.3d$. Second, for even $3d \leq N \leq 3.5d$,

$$|[t^N] \tan^{3d} t K_3(\tan t)| < 220(\log_2 d + 4)^2 2^d. \quad (5)$$

The latter follows from the orbit marking $\binom{L+1}{2}$ for H , where L is the deepest occupied dyadic factor. The complete proof is Lemma 2.1 of [3].

For the operator estimates, set

$$R = 2/5, \quad w = 49/100, \quad \mathcal{H}_a(x) = \prod_{j \geq 1} \left(\cos \frac{\pi x}{2^j} + a \sin \frac{\pi x}{2^j} \right),$$

$$h = \mathcal{H}_R, \quad L_R = h(1/2) = L(R).$$

On $X = \{f \in C^1(\mathbb{R}/\mathbb{Z}) : f(0) = 0\}$ use

$$N_0(f) = \sup_{x \notin \mathbb{Z}} \frac{|f(x)|}{t(x)h(t(x))^d}, \quad N_1(f) = \sup_x \frac{|f'(x)|}{h(t(x))^d},$$

$$\|f\|_d = N_0(f) + \frac{(9/10)^d}{d} N_1(f), \quad t(x) = \text{dist}(x, \mathbb{Z}).$$

For even $d \geq 128$ and $|a| \leq R$, the inverse of $I - QV_a|_X$ has norm at most five. Evaluation at $1/2$ costs $L_R^d/2$. Also $1 \leq h \leq 7/5$ on $[0, 1/2]$, h increases on $[0, 1/4]$, and each individual inverse-branch ratio satisfies

$$|W_a(y)|h(t(y))/h(t(x)) \leq 1, \quad W_a(y) = \cos \pi y + a \sin \pi y. \quad (6)$$

The normalized frozen function $f = h_0 + (I - QV_a)^{-1}q_a$ has the approximation

$$f = \phi - C + e, \quad \phi(x) = \mathcal{H}_a(x)^d + \mathcal{H}_a(x-1)^d, \quad (7)$$

$$\|C\|_\infty, \|C'\|_\infty \leq 610d(RL_R)^d, \quad \|e\|_d \leq 60000d^2w^d.$$

Here C is the cubic Hermite correction of the excess endpoint values and derivatives. Its scalar consequence is

$$|F(a) - L(a)^d - L(-a)^d| \leq 31000d^2(wL_R)^d, \quad wL_R < \beta := 11907/20000. \quad (8)$$

The same estimates give the full-disk bounds

$$|F(a)| \leq 3L_R^d, \quad |G(a)| \leq 4275dL_R^d. \quad (9)$$

The proof of the G bound uses $\|f - h_0\|_d \leq 1710d$ and the unchanged Green inverse; it does not extend an unregularized orbit sum beyond its convergence range.

Finally, writing $M_{d,q} = [a^q]B(a)^d$, the dyadic tangent comparison and low-degree orbit formula give

$$|F_q| \leq \frac{5}{2}c^d M_{d,q}, \quad |G_q| \leq (5 \log_2 d + 18)c^d M_{d,q} \quad (0 \leq q \leq d-2). \quad (10)$$

A monomial carries weight one for F and its deepest occupied index for G . Odd coefficients vanish. The tangent comparison and orbit identities are included in the preceding source archive.

3 The actual quadratic asymptotic

Write

$$A_k(d, N) = [t^N] \mathcal{G}(t)^{kd}.$$

Proposition 3.1. *Uniformly for even integers N with N/d in any compact subinterval of $(2, 2\mu(\arctan R))$,*

$$[t^N] \tan^{2d} t K_2(\tan t) = (2\log_2 d + O(1)) A_2(d, N). \quad (11)$$

Proof. We first quantify the deepest occupied index. For q/d in a compact subinterval of $(0, \infty)$, choose $a > 0$ so that

$$\nu(a) := \sum_j \frac{ab_j}{1 + ab_j} = q/d.$$

Take d independent labeled copies of each Bernoulli selector with probability $ab_j/(1 + ab_j)$. Conditioning their total count on q gives precisely the coefficient measure of $M_{d,q}$. Let L be the largest selected index. Uniformly in this parameter range,

$$\mathbb{E}L = \log_2 d + O(1). \quad (12)$$

Here is a direct uniform justification. The total count has mass at its integer mean at least $c_0/\sqrt{d}$, by the lattice local limit theorem. All parameters remain in a compact positive interval; there are common exponential moments and a uniform span-one gap. The maximum mass after removing one selector, or after retaining only indices up to $J \geq 1$, is at most $C_0/\sqrt{d}$: an independent binomial summand from the $j = 1$ selectors supplies this bound, and convolution cannot increase it. The selector tail is comparable to 2^{-J} . Marking one tail selector and, in the other direction, requiring all tail selectors to vanish, give

$$\mathbb{P}(L > J \mid q) \leq \min(1, Cd2^{-J}), \quad \mathbb{P}(L \leq J \mid q) \leq Ce^{-cd2^{-J}}.$$

Summation above and below $\lfloor \log_2 d \rfloor$ proves (12), uniformly.

For even integers q with q/d in a compact subinterval of $(0, 1)$, the two nearest half-integer orbits consequently give

$$F_q = 2c^d M_{d,q}(1 + O(e^{-c_1 d})), \quad G_q = 2c^d M_{d,q}(\log_2 d + O(1)). \quad (13)$$

Indeed the remote odd branches have an extra factor n^{q-d} , with at most logarithmic depth factors; their sums for $n \geq 3$ are exponentially small uniformly when $q \leq (1 - \epsilon)d$.

Let $F_<, G_<$ retain raw degrees below d . From (9), their high tails on $|a| \leq a_0 < R$ are bounded by

$$|F - F_<| \leq \frac{3L_R^d(a_0/R)^d}{1 - a_0/R}, \quad |G - G_<| \leq \frac{4275dL_R^d(a_0/R)^d}{1 - a_0/R}.$$

The low parts are bounded by a logarithmic factor times $L(a_0)^d$, using (10). Their relative tail factor is

$$\tau(a_0) = \frac{a_0 L_R}{R L(a_0)} < 1.$$

This follows because $a/L(a)$ increases on $(0, R]$: $\nu(a) < a \sum b_j < 2a \leq 4/5$. At the second saddle, $a_0 = \tan t_2$ stays in a compact subinterval of $(0, R)$. Cauchy's estimate and the local-limit lower bound for A_2 therefore make every product containing a high tail exponentially negligible.

In the remaining low convolution, mark the two raw selector counts by u, v in

$$c^{2d} \tan^{2d} t B(u \tan t)^d B(v \tan t)^d.$$

At the positive saddle each count has mean $d\nu(a_0)$, lying in a compact subinterval of $(0, 1)$. Conditional on total t -degree N , either count outside a fixed smaller interior interval has

exponentially small probability. This follows by marking with e^θ and a Chernoff bound at the same saddle; division by the local-limit coefficient costs only $O(\sqrt{d})$. The absolute bounds (10) control these atypical terms. Thus (13) applies on the whole convolution up to an $O(A_2)$ error.

Set $F_{\text{near}}(a) = c^d(B(a)^d + B(-a)^d)$. Restoring its full nearest product instead of its truncation adds only the same atypical raw-degree tail. We obtain

$$[t^N] \tan^{2d} t F_{<} G_{<} = \log_2 d [t^N] \tan^{2d} t F_{\text{near}}(\tan t)^2 + O(A_2).$$

For even d, N , the last coefficient equals

$$2A_2(d, N) + 2[t^N] \{\mathcal{G}(t)\mathcal{G}(-t)\}^d.$$

The mixed term is exponentially negligible. On $|z| = t_2$ its product modulus is strictly below $\mathcal{G}(t_2)^2$, uniformly on the compact saddle interval. Equality for the first factor would require $z/t_2 = 1$, while equality for the second would require $z/t_2 = -1$, since consecutive coefficients are positive. Finally $F_{<}^2/2$ contributes only $O(A_2)$. This proves (11). $\square$

4 A bounded local frozen correction

The first-model saddle lies outside the Green disk. We do not approximate the actual frozen function there. Instead, subtract its explicit nearest model and estimate the remainder on the smaller second-saddle contour.

Let

$$I = [3.3, 3.332], \quad t_- = .3413848, \quad t_+ = .3484140, \quad \theta_0 = 153\pi/1000.$$

The rational saddle certificate gives $t_2(\sigma) \in [t_-, t_+]$ for $\sigma \in I$. On the compact local arcs

$$a = \tan(te^{i\theta}), \quad t_- \leq t \leq t_+, \quad |\theta| \leq \theta_0, \quad (14)$$

the exact local certificate proves, with strict slack,

$$\Re a > .303, \quad .348 < |a| < .38, \quad |Q_a(2^{-j})| < 1 \quad (1 \leq j \leq 5), \quad Q_a(x) = \frac{\mathcal{H}_a(1+x)}{\mathcal{H}_a(1)\mathcal{H}_a(x)}. \quad (15)$$

A common strict upper bound below one is supplied for the five ratios. The certificate and its interval construction are described in Section 5.

Put $L_a = \mathcal{H}_a(1/2) = L(a)$ and $g_a = aL_a$. The following lemma is the local analytic ingredient.

Lemma 4.1. *Uniformly on the arcs (14),*

$$F(a) = L(a)^d + L(-a)^d - a^d L(a)^{2d} \{\Psi_d(a) + O(d^{-1})\}, \quad (16)$$

where Ψ_d is analytic on a common neighborhood and uniformly bounded there. In particular,

$$|F(a) - L(a)^d - L(-a)^d| \leq C|aL(a)^2|^d.$$

Proof. We give the principal-branch identity and all error estimates. The atomic function has the periodization

$$h_0(x) = \sum_{k \in \mathbb{Z}} \left(\frac{\sin \pi(x+k)}{\pi(x+k)} \right)^d.$$

To distinguish its nonperiodic summand, write $s(x) = \sin(\pi x)/(\pi x)$. The frozen equation is $f = V_a f - a^d F h_0$. Divide by $\mathcal{H}_a(x)^d$ and use $\mathcal{H}_a(x) = W_a(x/2)\mathcal{H}_a(x/2)$. If $x_n = 2^{-n-1}$, telescoping toward zero gives the exact, convergent identity

$$\frac{F}{L_a^d} = 1 + \sum_{n \geq 0} \frac{W_a((1+x_n)/2)^d f((1+x_n)/2) - a^d F h_0(x_n)}{\mathcal{H}_a(x_n)^d}. \quad (17)$$

Each numerator vanishes at zero before it is summed.

The weighted error $f - \phi$ and its derivative are bounded by $\text{poly}(d)w^d h(t(x))^d$. For the derivative this needs a small refinement of (7), rather than division by its exponentially small norm weight. The two-row Green estimate gives

$$N_1(e) \leq 5dN_0(e) + \beta_d N_1(e) + N_1(\text{source}), \quad \beta_d < 3/5,$$

with $N_0(e) = O(d^2 w^d)$ and $N_1(\text{source}) = O(d^2 w^d)$. Hence $N_1(e) = O(d^3 w^d)$, and the Hermite correction has the same exponential base.

For $0 \leq x \leq 1/2$, factorwise absolute values give $|\mathcal{H}_a(x)| \geq \mathcal{H}_{\Re a}(x) \geq 1$. The last inequality follows from real logarithmic concavity, the value one at zero, and the certified $L(.303) > 1.05$. Thus $|g_a| > .348 \cdot 1.05 = .3654$. The squared secondary weight $|a \cos(\pi x/2) - \sin(\pi x/2)|^2$ has its maximum at an endpoint; its derivative changes sign only from negative to positive. By (15), its endpoint bounds, multiplied by $h \leq 7/5$, are below .73. Substituting ϕ for f in the secondary branch of (17), and $F_{\text{near}} = L(a)^d + L(-a)^d$ for F , therefore has normalized error

$$O(\text{poly}(d)(.73w)^d), \quad .73w = .3577 < |g_a|.$$

The rank-one error has smaller base $|a|\beta$. These error numerators still vanish at zero. Their normalized derivatives obey the same bound up to a polynomial factor, so summing the geometric sequence x_n is legitimate.

The substituted secondary product is exactly

$$W_a((1+x)/2)^d \phi((1+x)/2) = \mathcal{H}_a(1+x)^d + \mathcal{H}_a(x-1)^d.$$

Replacing $h_0(x)$ by $s(x)^d$ has exponentially smaller cost: the remaining periodization sum is $O((2/\pi)^d)$, with derivative $O(d(2/\pi)^d)$. Their vanishing at zero again controls the dyadic sum.

The negative-branch term at $n = 0$ equals $L(-a)^d/L(a)^d$, recovering the second term of F_{near} . Its counterterm is retained in the exponentially small remainder. For $n \geq 1$, $0 \leq x_n \leq 1/4$ and $|\mathcal{H}_a(x_n - 1)| \leq |a|$: the first factor satisfies this because $\tan(\pi/8) < 2\Re a/(1 - |a|^2)$, and every later factor has modulus at most one. Also $|L(-a)| < 1$. The difference at zero vanishes, and its derivative has base $|a|^d$, exponentially smaller than $|g_a|^d$. The removed counterterm has the still smaller relative base $c|L(-a)|/|L(a)|^2 < 1$.

It remains to analyze the positive branch. Put

$$v_a(x) = \frac{s(x)}{\mathcal{H}_a(x)}, \quad J(a) = \pi a - \frac{\mathcal{H}'_a(1)}{\mathcal{H}_a(1)}.$$

The preceding estimates reduce (17) to

$$\frac{F_{\text{near}} - F}{a^d L(a)^{2d}} = \sum_{n \geq 0} \{v_a(x_n)^d - Q_a(x_n)^d\} + O(e^{-c_1 d}). \quad (18)$$

All constants are uniform on a neighborhood of the compact arcs, by the strict certificate margins.

We have $\Re J(a) > \pi \Re a > .9$. Indeed, for real $u \geq 0$,

$$\Re \frac{a - u}{1 + au} \leq \Re a,$$

because $\Re((1 + a^2)/(1 + au)) > 0$ when $\Re a > 0$ and $|a| < 1$. Summing after the first spatial factor gives $\Re(\mathcal{H}'_a(1)/\mathcal{H}_a(1)) \leq \pi(\Re a - \Re(1/a))/2 < 0$. For $0 \leq x \leq 1/64$, the real part of the first factor of $\mathcal{H}_a(1+x)$ exceeds .27; the other real parts exceed $2/3$. All factor real parts of $\mathcal{H}_a(x)$ exceed .99. Therefore

$$|(\log Q_a)''(x)| \leq \pi^2(1 + .38^2) \left\{ \frac{1}{4(.27)^2} + \frac{3}{16} + \frac{1}{3(.99)^2} \right\} < 46.$$

Since $(\log Q_a)'(0) = -J(a)$, this proves $|Q_a(x)| \leq e^{-x/2}$ on that interval. The five larger dyadic arguments are covered by (15). Also

$$v_a(x) = \prod_{j \geq 1} (1 + a \tan(\pi x/2^j))^{-1}$$

has a uniform bound $|v_a(x)| \leq e^{-c_2 x}$ on $[0, 1/2]$.

Uniform Taylor expansion now gives

$$\log v_a(x) = -\pi a x + O(x^2), \quad \log Q_a(x) = -J(a)x + O(x^2).$$

Replacing the powers in (18) by these exponentials costs $O(1/d)$: each error is at most $C dx_n^2 e^{-c_3 dx_n}$, and the sum of $(dx_n)^2 e^{-c_3 dx_n}$ is uniformly bounded. The finitely many larger arguments cost exponentially little. Extending the resulting sum to all integer indices also costs exponentially little. Thus (16) holds with

$$\Psi_d(a) = \sum_{n \in \mathbb{Z}} \left\{ e^{-\pi a d 2^{-n-1}} - e^{-J(a) d 2^{-n-1}} \right\}. \quad (19)$$

The positive real parts of πa and $J(a)$ make this series uniformly convergent and bounded, with any fixed number of derivatives, on a common neighborhood. This completes the proof. $\square$

For real a in the corresponding interval, $J(a) > \pi a$ and $\Psi_d(a) > 0$. Moreover $\Psi_{2d}(a) = \Psi_d(a)$, and its mean over one period in $\log_2 d$ is $\log_2(J(a)/(\pi a))$, by Frullani's integral. Only its uniform boundedness is needed for the first-negative law.

5 The outer contour and its exact certificate

For $0 < |a| \leq R$, define

$$\omega(a) = \sup_{1 \leq X \leq 2} \frac{\max(|\mathcal{H}_a(X)|, |\mathcal{H}_{-a}(X)|)}{h(\text{dist}(X, \mathbb{Z}))}.$$

Lemma 5.1. *On any compact subset of $0 < |a| \leq R$,*

$$|F(a) - L(a)^d - L(-a)^d| \leq C d^2 \{L_R \omega(a)\}^d. \quad (20)$$

Proof. The continuous function ω has a positive lower bound on the compact set: its endpoint includes $|aL(a)|$, and $L(a) = cB(a)$ has no zero for $|a| < 1$. The nearest-pair excess endpoint values are bounded by ω^d , and its excess endpoint derivatives by $120d\omega^{d-1}$. Thus the fixed Hermite basis gives $\|C\|_\infty + \|C'\|_\infty \leq C_1 d \omega^d$.

For $e = f - (\phi - C)$ the periodic vanishing source is

$$s = \mathcal{H}_a(x-2)^d + \mathcal{H}_a(x+1)^d - (V_a C - C) - a^d(\phi - C)(1/2)h_0, \quad s(0) = 0.$$

The outer derivative satisfies

$$\frac{|(\mathcal{H}_a(X)^d)'|}{h(\text{dist}(X, \mathbb{Z}))^d} \leq 120d\omega^{d-1} \leq C_2 d \omega^d.$$

For the transfer of C , (6) and $h \geq 1$ imply both $|W_a(y)|^d/h(t(x))^d \leq 1$ and $|W_a(y)|^{d-1}/h(t(x))^d \leq 1$. Differentiation therefore gives $N_1(V_a C) \leq C_3 d^2 \omega^d$. For the rank-one term, $|a|^d |\mathcal{H}_a(\pm 1/2)|^d \leq \omega^d$ by the endpoint identity; the correction involving C is no larger. Since $\|h'_0\| \leq \pi d$, its derivative has the same bound. Hence $N_1(s) \leq C_4 d^2 \omega^d$.

For $t(x) \leq 1/4$, integrate from the nearest integer using $s(0) = 0$ and increasing h . For $t(x) \geq 1/4$, bound the weighted amplitudes directly and divide by $t(x)$. This yields $\|s\|_d \leq C_5 d^2 \omega^d$. Apply the norm-five Green inverse, evaluate at $1/2$, and add the Hermite value correction. This proves (20) with its exact exponential base. $\square$

The outer certificate proves, with $\eta = 1999/2000$, that

$$|a|L_R \frac{\max(|\mathcal{H}_a(X)|, |\mathcal{H}_{-a}(X)|)}{h(\text{dist}(X, \mathbb{Z}))} < \eta g(a_0)^2, \quad (21)$$

for

$$t_- \leq t \leq t_+, \quad \theta_0 \leq \theta \leq \pi/2, \quad 1 \leq X \leq 2, \quad a = \tan(te^{i\theta}), \quad a_0 = \tan t.$$

Conjugation and evenness cover the rest of the circle.

For completeness, the certificate is a finite rational inequality proof, not a floating-point search. The complex tangent is enclosed by

$$\tan(u + iv) = \frac{\sin(2u) + i \sinh(2v)}{\cos(2u) + \cosh(2v)}, \quad u = t \cos \theta, \quad v = t \sin \theta.$$

Machin's formula and Taylor remainder bounds provide outward integer intervals for all trigonometric and hyperbolic functions. Twenty spatial factors retain the same complex interval a . For $c_j = \cos(\pi X/2^j)$ and $s_j = \sin(\pi X/2^j)$, the exact squared factor is

$$|c_j \pm as_j|^2 = c_j^2 \pm 2\Re(a)c_j s_j + |a|^2 s_j^2.$$

Its logarithmic derivative is

$$2 \frac{\pi \pm \Re(a)(c_j^2 - s_j^2) + (|a|^2 - 1)c_j s_j}{2^j |c_j \pm as_j|^2}.$$

For the omitted product, set $E = R\pi X_{\max} 2^{-20} + (\pi X_{\max})^2 / (6 \cdot 4^{20})$. The sum of the omitted factor deviations is at most $E < 1$, so its modulus is at most $(1 - E)^{-1}$. The omitted squared logarithmic derivative is bounded by

$$\frac{2\pi}{2^{20}} \frac{R + \tau}{1 - R\tau - \tau^2/2}, \quad \tau = \frac{\pi X_{\max}}{2^{21}}.$$

Spatial cells are split at $X = 3/2$. For the real denominator, positivity and logarithmic concavity make its minimum on a distance cell occur at an endpoint. A midpoint value and an interval bound for the logarithmic derivative control a spatial box by $e^v \leq 1/(1 - v)$ when $0 \leq v < 1$. A direct squared-factor bound is used whenever that sharpening is unavailable.

The compact outer partition has 2812 boxes. Every squared ratio is below $999/1000 < (1999/2000)^2$; its largest certified upper bound is approximately .9989983569. The local partition has 127 boxes and largest certified shallow Q squared upper bound

$$\frac{99983441723839698176893318289531}{10^{32}} < 1.$$

The box partitions, exact rational endpoints, interval programs, and independent replay checks are included. The algorithm starts from the whole domain and bisects a box into two children; replay reconstructs the partition and checks every leaf. All new scalar constants, saddle coverage, and the fact that both model exponential bases exceed three on I are also checked exactly. Strict margins give the compact-neighborhood slack used in Lemma 4.1.

6 The coefficient comparison and the first negative degree

Proposition 6.1. *Uniformly for even N with $N/d \in I$,*

$$[t^N]P_m(t) = 2A_1(d, N) - (2 \log_2 d + O(1))A_2(d, N). \quad (22)$$

Proof. Let $t = t_2(N/d)$. On the positive local arc, Lemma 4.1 bounds the composed frozen remainder by $C|\mathcal{G}(z)|^{2d}$. For the normalized positive coefficient law at radius t , with probabilities p_j , one has

$$\left| \sum_j p_j e^{ij\theta} \right|^2 \leq 1 - 2p_1 p_2 (1 - \cos \theta).$$

The two probabilities are uniformly positive, so this gives a uniform Gaussian modulus bound. Integration over the arc retains the factor $d^{-1/2}$. The negative local arc has the same bound by evenness. On the outer arcs, Lemma 5.1 and (21) give an exponentially smaller term, even after division by the second-model local-limit coefficient. Thus

$$[t^N] \tan^d t F(\tan t) = 2A_1(d, N) + O(A_2(d, N)).$$

The exact factor two comes from the equal even coefficients of $\mathcal{G}(t)^d$ and $\mathcal{G}(-t)^d$.

Proposition 3.1 handles the quadratic term. By (5), the cubic contribution is polynomial in $\log d$ times 2^d . Both model exponential bases exceed three uniformly on I , as certified, so this is exponentially negligible relative to A_2 . The cosine term is $O(d)$ by Cauchy's estimate on $|t| = 1$. The exact expansion (4), valid since $N < 4d$, proves (22). $\square$

Proof of Theorem 1.1. Uniform positive-series saddle asymptotics give

$$\frac{A_2(d, N)}{A_1(d, N)} = C(\sigma) e^{d\Delta(\sigma)} (1 + o(1)), \quad C(\sigma) = \sqrt{\frac{v_1(\sigma)}{2v_2(\sigma)}}, \quad v_k = t_k \mu'(t_k), \quad \sigma = N/d. \quad (23)$$

The errors are uniform on I . The local-limit proof uses a common analytic neighborhood, positive variance bounds, and the uniform span-one gap from consecutive positive coefficients. In particular C is bounded above and below by positive constants.

Since Δ increases and vanishes at σ_* , (22) is positive uniformly below $\sigma_* - \epsilon$ and negative at $\sigma_* + \epsilon$, for each sufficiently small fixed $\epsilon > 0$ and all large d . The preceding eventual-6.6m theorem covers every earlier degree to the left of I . Therefore $N_m/d \rightarrow \sigma_*$.

For the sharper location, put

$$M_d = d\sigma_* - \frac{\log \log d}{\Lambda}.$$

Uniformly when $N - M_d$ is bounded,

$$d\Delta(N/d) = -\log \log d + \Lambda(N - M_d) + o(1).$$

Choose constants $0 < C_{\min} \leq C(\sigma) \leq C_{\max}$ on I . Since Δ increases, every degree below $M_d - K$ has

$$\log_2 d \frac{A_2}{A_1} \leq \frac{C_{\max}}{\log 2} e^{-\Lambda K + o(1)}.$$

A sufficiently large fixed K makes this less than $1/2$, proving positivity at all such degrees within I . The earlier theorem covers the rest. At an even degree in $[M_d + K, M_d + K + 2]$, the corresponding lower bound using $C_{\min}$ exceeds two, giving a negative coefficient. No monotonicity of the full coefficient ratio is assumed. We conclude $N_m = M_d + O(1)$, which is (3). $\square$

7 A refined lattice window

Let $v_k = t_k \mu'(t_k)$ at the crossing, and put

$$C_* = \sqrt{v_1/(2v_2)}, \quad u_d = d\sigma_* - \frac{\log \log d}{\Lambda} + \frac{\log((\log 2)/C_*)}{\Lambda}.$$

Proposition 7.1. *There are constants C, B, d_0 such that, for even $d \geq d_0$, every even post-cancellation degree $N < u_d - C/\log d$ has positive pressure coefficient, while every even degree with*

$$u_d + C/\log d < N \leq u_d + B$$

has negative coefficient. In particular, whenever $\text{dist}(u_d, 2\mathbb{Z}) > C/\log d$,

$$N_m = 2 \left\lceil \frac{u_d}{2} \right\rceil.$$

Proof. The compact family of positive saddle laws gives the quantitative version of (23), with relative error $O(d^{-1/2})$. Indeed, on a fixed sufficiently small central arc, the third-order characteristic expansion bounds the integral error by

$$O\left(d \int |\theta|^3 e^{-cd\theta^2} d\theta\right) = O(d^{-1}),$$

against a main integral of order $d^{-1/2}$. The complement has a uniform exponential gap. This makes the error uniform in the compact slope interval.

For fixed B and $|N - u_d| \leq B$, Taylor expansion at σ_* and (22) therefore give

$$\frac{[t^N]P_m(t)}{2A_1(d, N)} = 1 - e^{\Lambda(N-u_d)} + O_B(1/\log d).$$

The all-earlier-sign argument already gives positivity below $u_d - B_0$ for some fixed B_0 . Choose $B > \max(B_0, 3)$. On $[-B, B]$, the derivative of $e^{\Lambda x}$ is bounded below by a positive constant. A sufficiently large fixed C gives the two stated sign bounds. If the center is farther than $C/\log d$ from every even integer, the next even integer is negative and every earlier post-cancellation degree is positive, proving the last assertion. $\square$

The constants are existential. No claim is made about how often u_d approaches the even lattice, so this is not an unconditional exact floor rule.

8 Scope and reproducibility

The first-negative asymptotic is proved by the analytic comparison and exact compact-domain certificates. It is not inferred from a finite fit. Its slope is strictly below $20/3$, so a floor rule with that limiting slope cannot hold eventually. No unconditional rounding formula or effective onset threshold is supplied.

The real frozen correction contains a bounded oscillation in $\log_2 d$. Its contribution is smaller than the quadratic logarithmic multiplier, which is why it does not alter the leading slope or logarithmic correction. Proposition 7.1 isolates an $O(1/\log d)$ ambiguity window around the even lattice; it does not determine how often that window is encountered.

The source archive contains the compact local and outer partitions, portable exact checkers, model-crossing certificate, source hashes, and unchanged analytic prerequisites. The supplied first-negative finite data, when distributed as a companion, are supplementary and are not a premise of the theorem.

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